Platonic and Heraclitean Language Change

Chasing the S-curve through Community Structure

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Question

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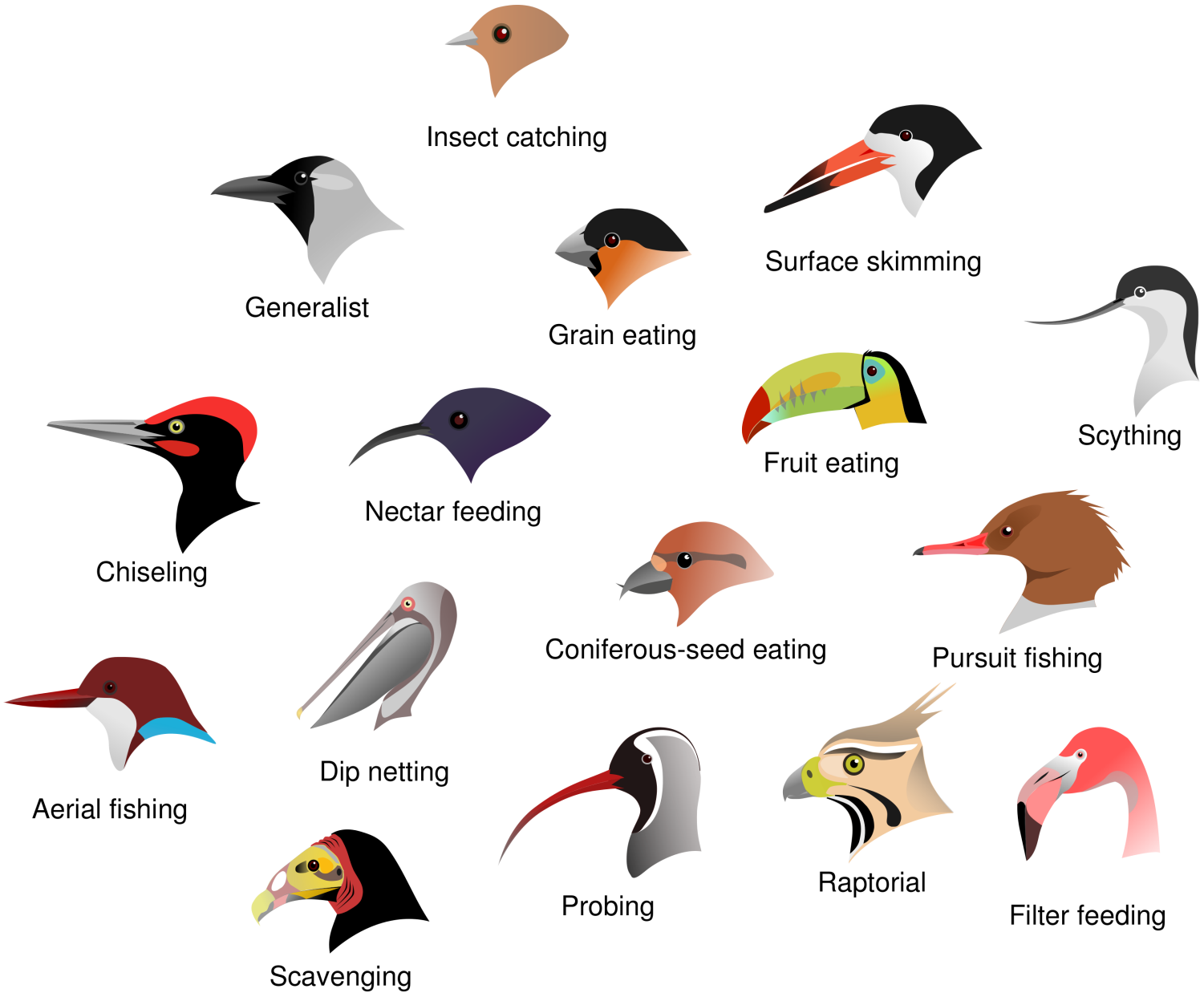
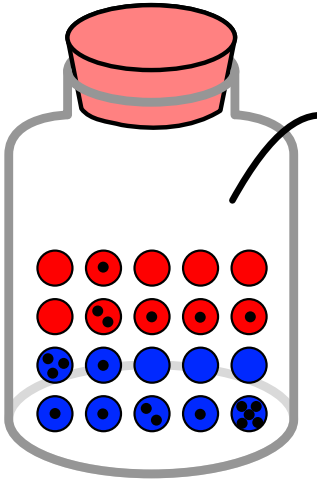
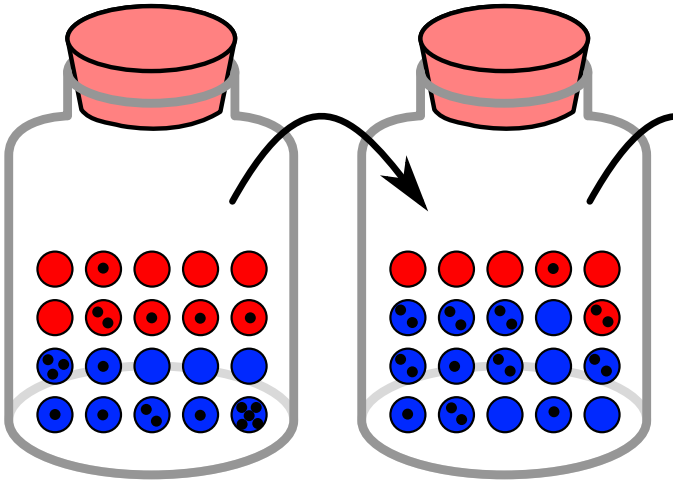
Is it possible to have language change that is **neutral**?

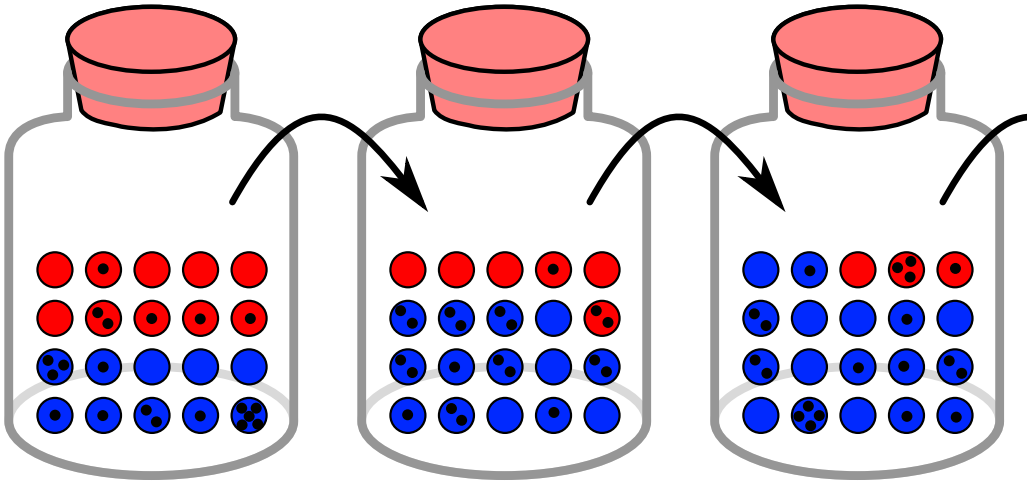
- Take two **competing** variants (grammars, parameters, allophones in free variation, etc.)
- If neither has selective advantage over the other (is not "fitter"), then model is neutral
- No room for prophylaxis, therapy, the Transparency Principle, third-factor processing considerations, prestige

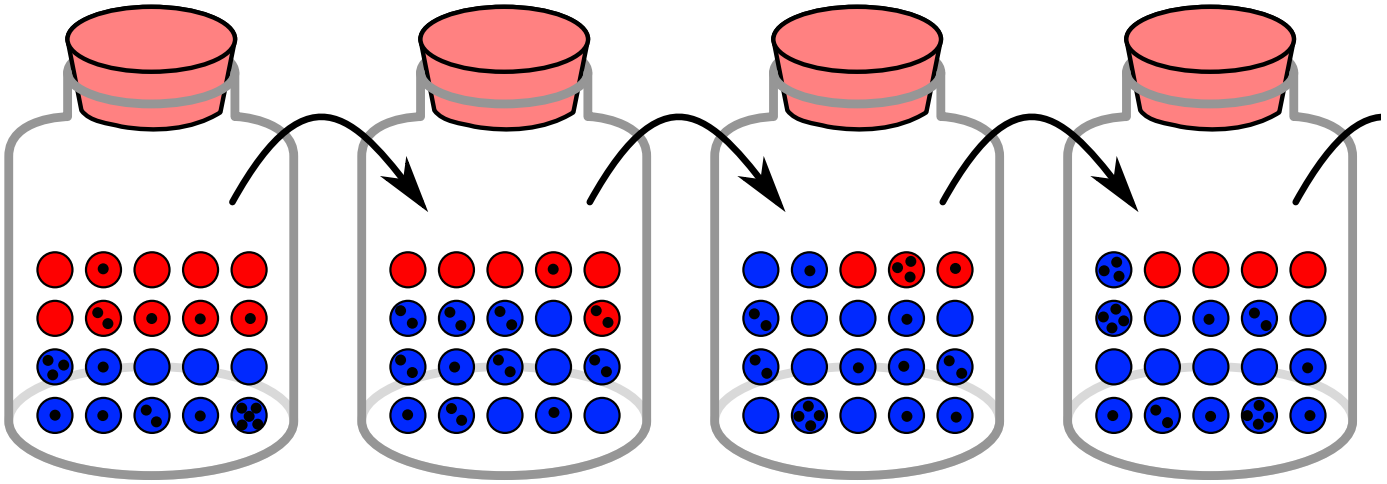
N.B. Is it **possible**, rather than is it **probable**…

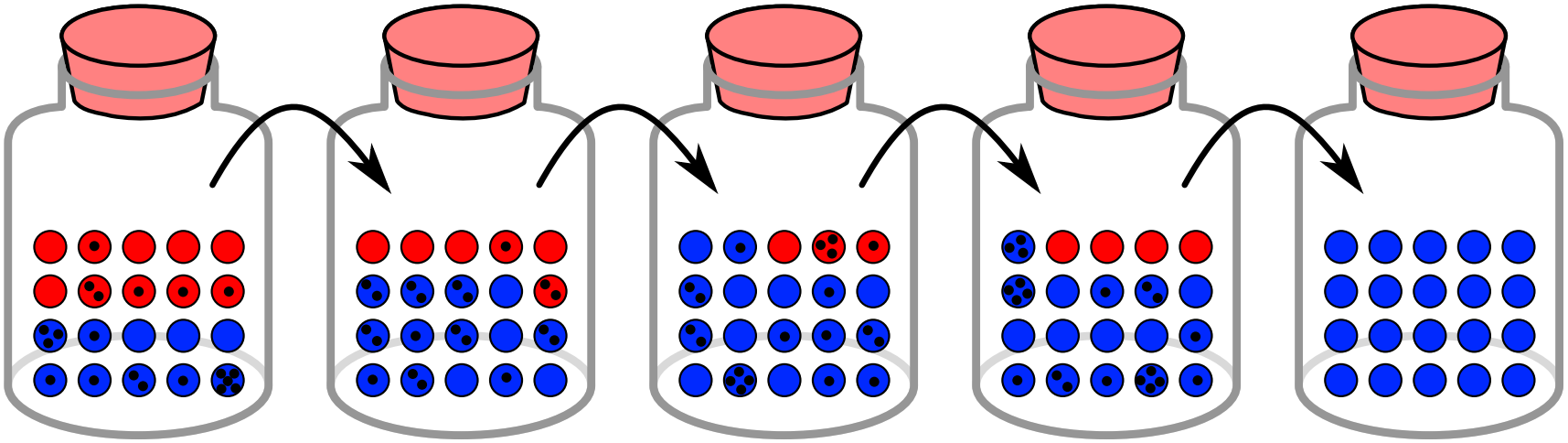

Insect catching
Generalist
Grain eating
Surface skimming
Scything
Fruit eating
Nectar feeding
Chiseling
Coniferous-seed eating
Pursuit fishing
Dip netting
Aerial fishing
Probing
Raptorial
Filter feeding
Scavenging











Answers

- So, **is** it possible to have language change that is neutral?
  - Ke et al. (2008): No. Only biased variants can spread reliably.
  - Fagyal et al. (2010): No. Only variants of prestigious and central speakers can spread along S-curves.
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- These are interesting mathematical/computational models of change.
- But: each assumes that **human social networks are static**.

?

What might happen in a dynamic, evolving network?

Explication

Neutral change

Acquisition/adoption of variant only depends on the frequency of that variant in the speaker's neighbourhood.

More restrictively: probability to adopt variant k equals the frequency of variant k in the speaker's neighbourhood.

Mathematically: the probability of adopting variant k is

$$p_k = \frac{1}{|\eta|} \sum_{i \in \eta} \chi_k(i), \tag{1}$$

where η is your neighbourhood and $\chi_k(i) = 1$ if speaker i has variant k and $= 0$ otherwise.

Mutations

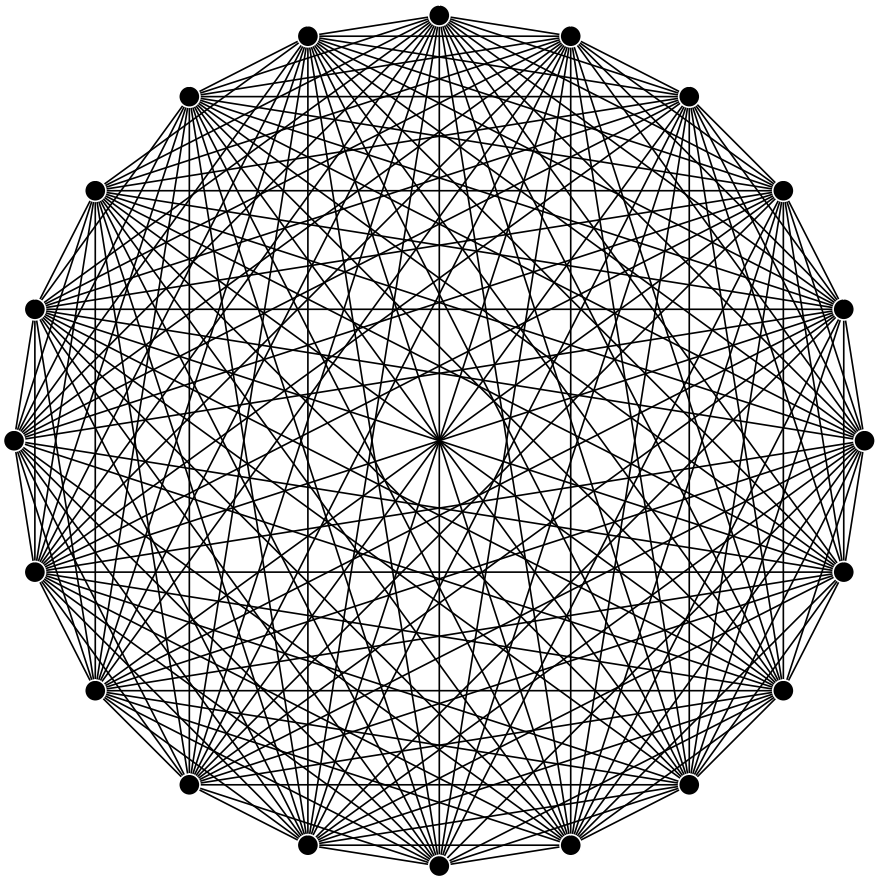
- But there is natural variation in adoption
- We replace (1) with (2):

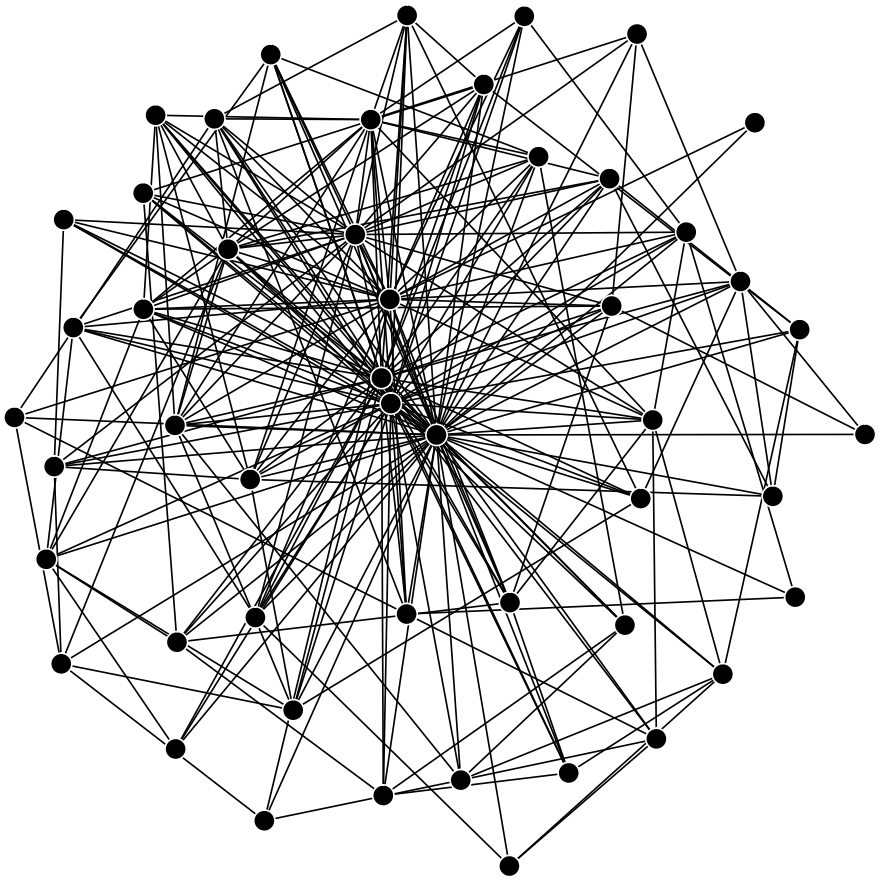
$$p_k = \frac{\mu}{C} + \frac{1-\mu}{|\eta|} \sum_{i \in \eta} \chi_k(i), \tag{2}$$

  where $0 \leq \mu \leq 1$ is a (usually small) **mutation** parameter and C is the number of competing variants.
- ICBS $\sum_{k=1}^{C} p_k = 1$, so this is well-defined.

(video: brown.mp4)

Neighbourhoods





Rewiring

- At each iteration, remove one speaker, add a new one, giving them K connections as follows: for each connection,
  - with probability σ, the connection is made to speakers who are the best connected
  - with probability $1 - \sigma$, the connection is made uniformly at random

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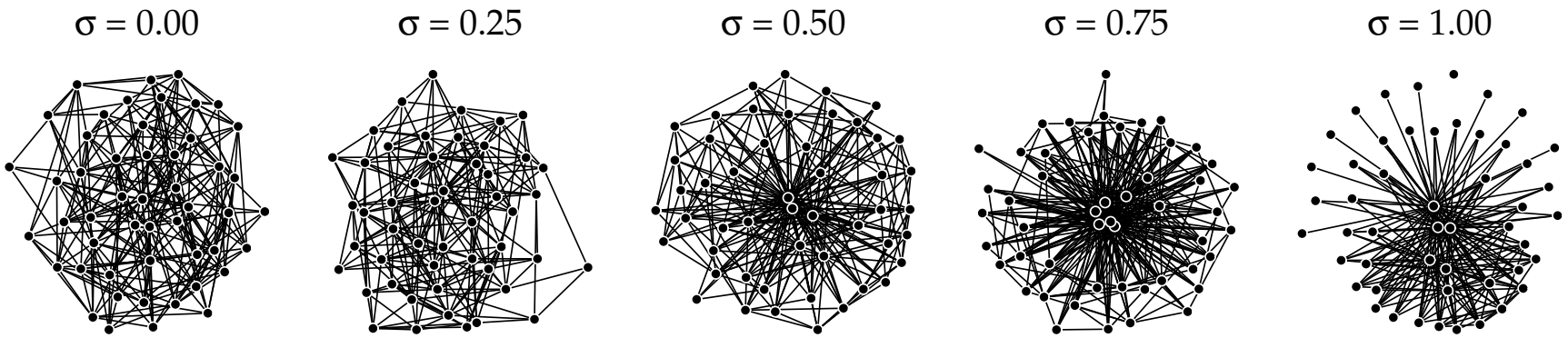
- Higher σ implies more clusterization



(videos: gr_mu00.mp4, gr_mu09.mp4)

Synopsis

- So, model has the following assumptions:
  - N speakers arranged in a social network
  - C competing linguistic variants
  - network rewired over time by removal and addition of speakers
  - this rewiring controlled by a preferentiality parameter σ
  - each new speaker given $K < N$ connections
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> “
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> *Idealization*, it should be noted, is a misleading term for the only reasonable way to approach a grasp of reality. (Chomsky, 1995, 7)

Desiderata

- Any model of language change should have the following properties (I claim), as language change has them (I claim):
  - **Dominance** (D): barring brief periods of change, a language community tends to relax into a state where a single variant dominates (has frequency $> 1 - \delta$ for suitable $\delta \ll 1$)
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  - **Well-behavedness** (W): these shifts are not too erratic, but follow something like monotone S-curves
- A model/reality that has these features I shall call **Platonic**
- A model/reality that violates (D) and/or (W) is **Heraclitean**
- A model/reality that violates (S) is **uninteresting**

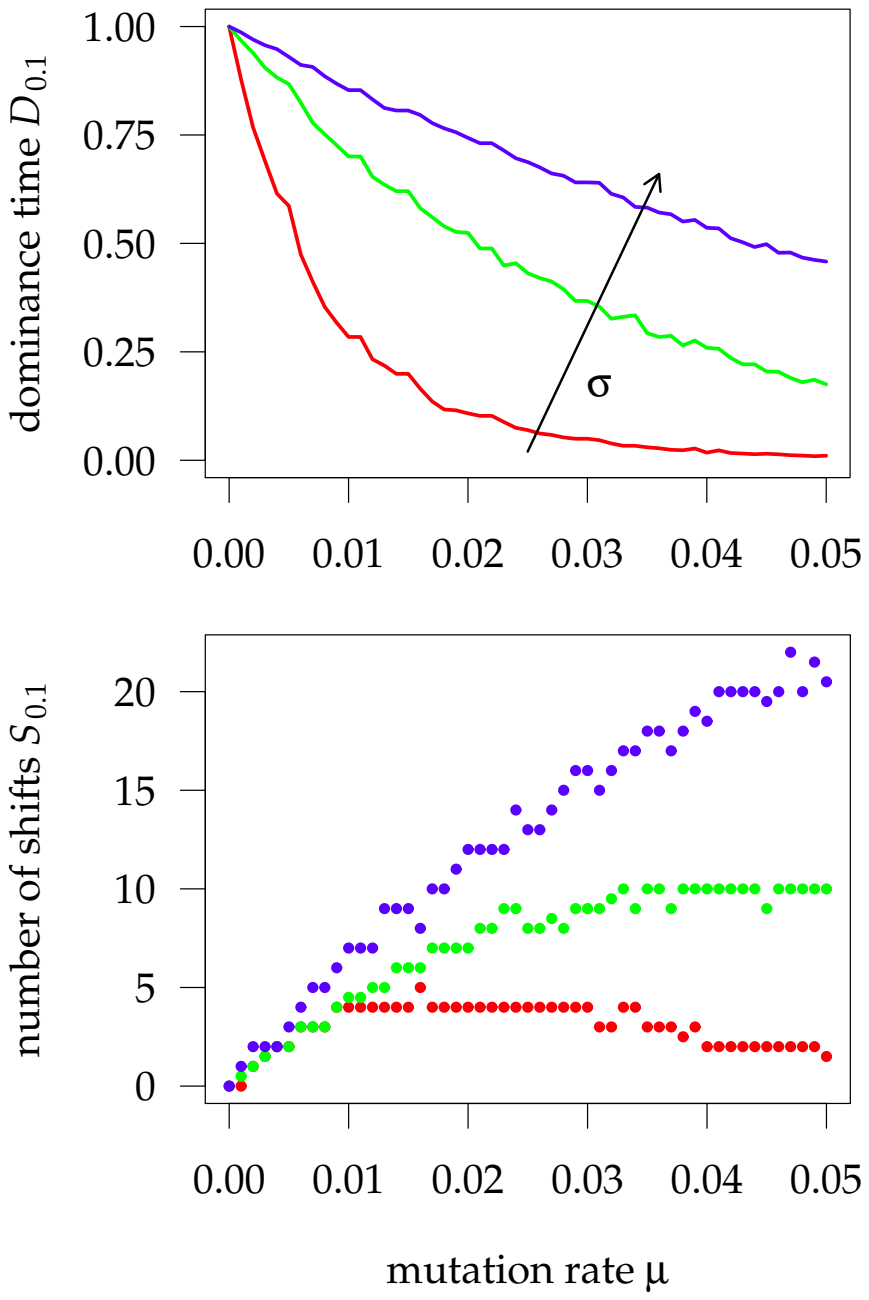
Evaluation

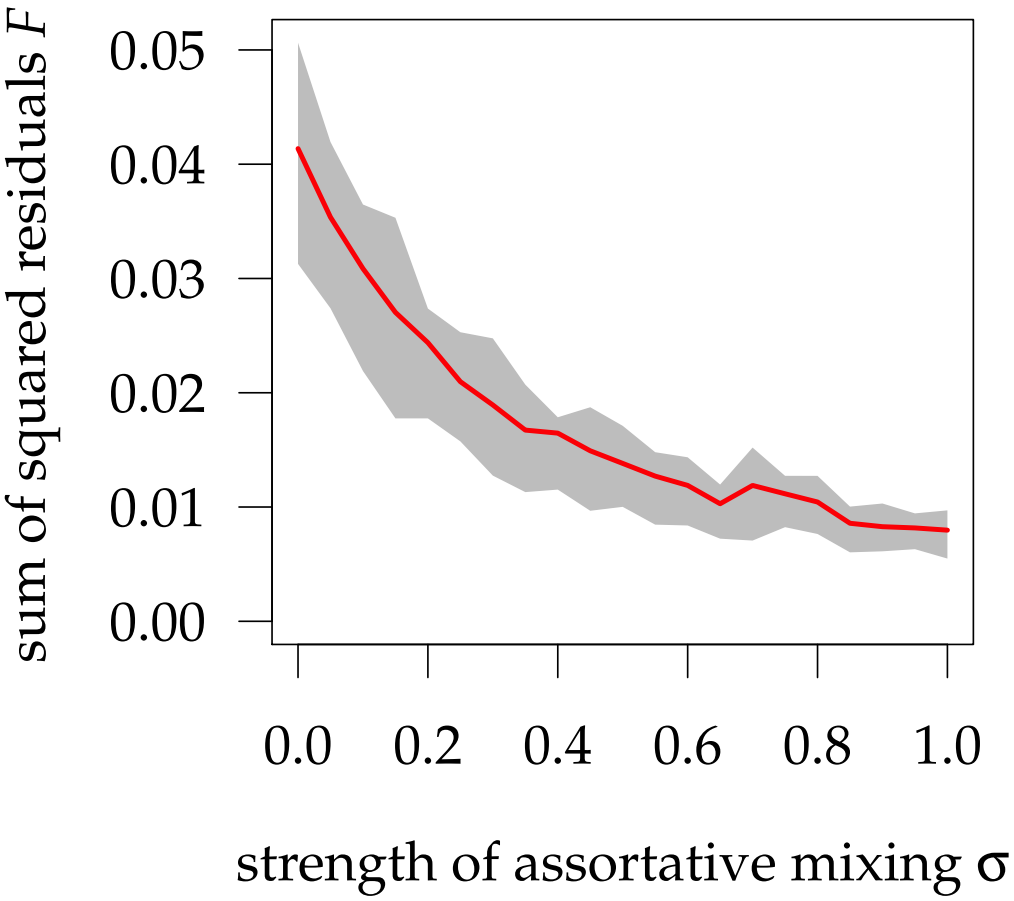
?

To what extent does model fulfil (D), (S) and (W)?

- Can look at individual simulation trajectories
- Can also define measures of (D), (S) and (W) and calculate these over a large batch of simulations
  - How does variation in model parameters (σ, μ) affect trajectories?
- Analytical results in limiting cases (at least $\sigma = 1$)
  - For large N, simulations computationally intractable
  - Understanding more valuable than simulating

(videos: ke1*.mp4, ke3*.mp4)

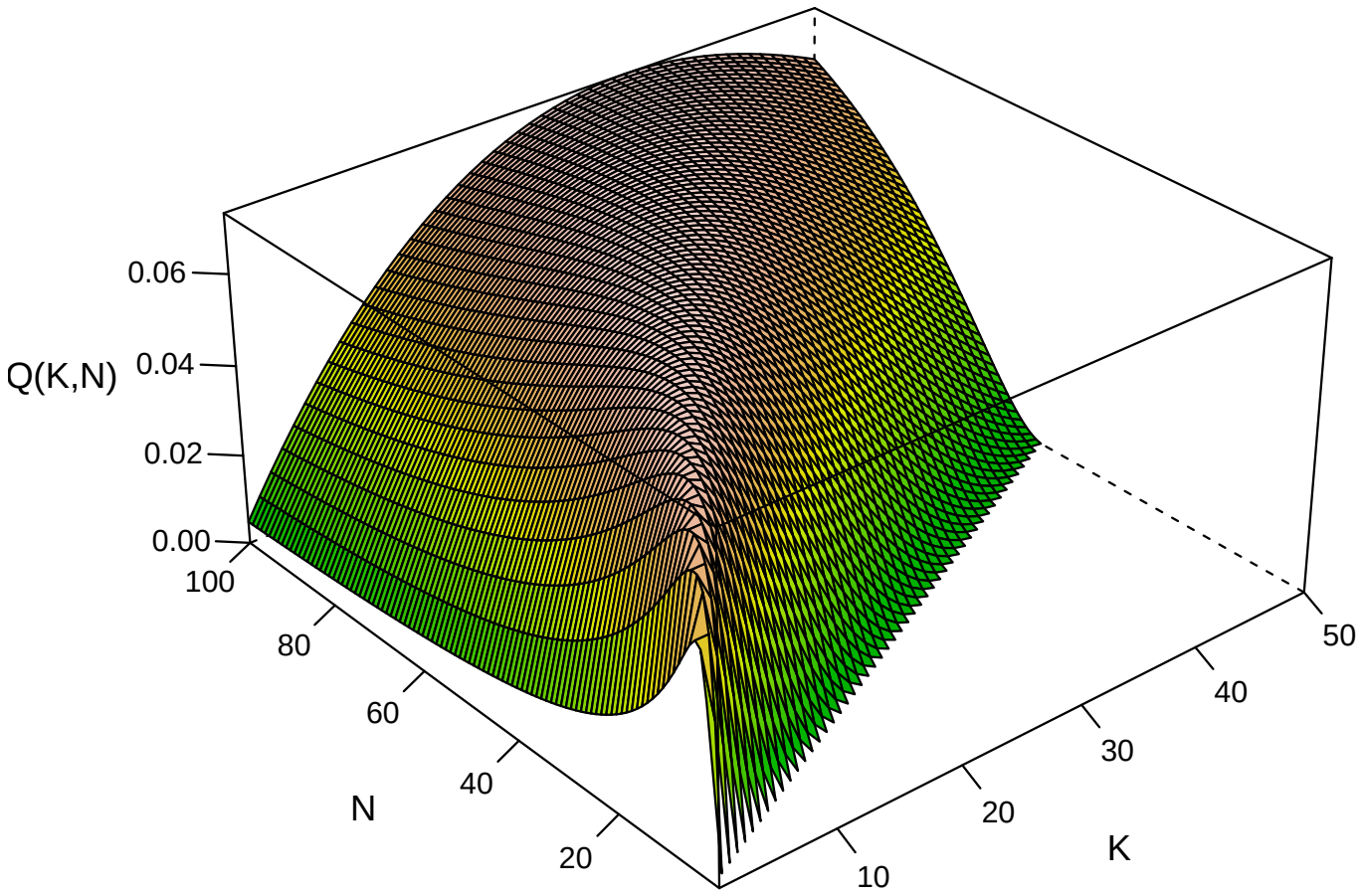

dominance time $D_{0.1}$
1.00
0.75
0.50
0.25
0.00
σ
0.00
0.01
0.02
0.03
0.04
0.05
number of shifts $S_{0.1}$
20
15
10
5
0
0.00
0.01
0.02
0.03
0.04
0.05
mutation rate μ



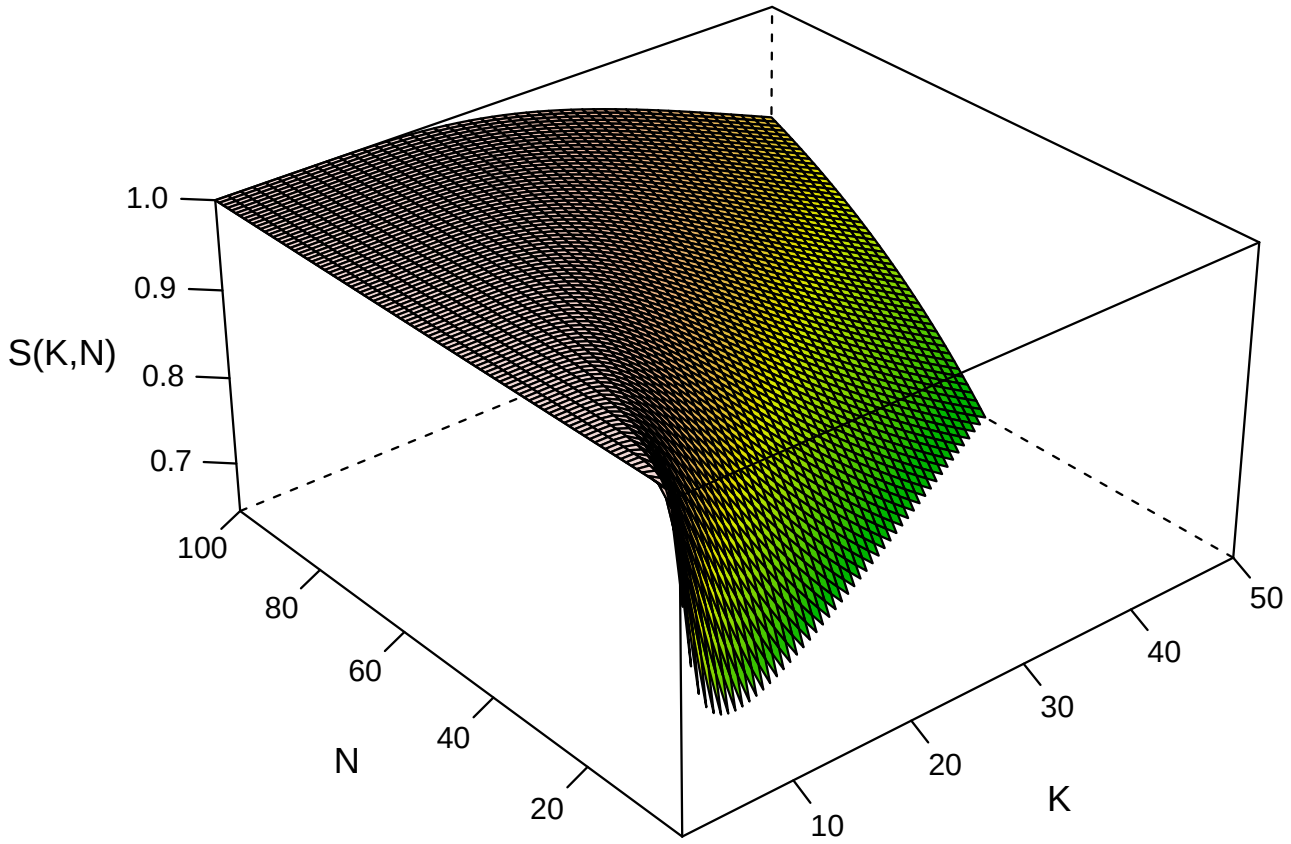
$$\text{mean displacement } M(\tau) = \frac{1}{n-\tau} \sum_{t=1}^{n-\tau} d(\vec{x}(t), \vec{x}(t+\tau))$$

Analytical results ($\sigma = 1$)

- Key insight: at any point of time, the speakers in the network may be partitioned as follows:
  - central speakers: degree $> K$
  - fringe speakers: degree $= K$
  - peripheral speakers: degree $< K$
- Can estimate probability of "promotion" from fringe to centre
- Can estimate probability that innovation propagates
- Additional insights:
  - innovations i.e. mutations occur in fringe
  - the more tight-knit the centre is, the more reliably an innovation propagates



$$Q(K,N) = \frac{(N-K)(N-K-1)K}{2N^3}$$



$$S(K,N) = 1 - \left(\frac{K-1}{K}\right)^N \quad (\text{always } S(K,N) > 1 - e^{-1} \approx 0.6)$$

“

So it’s perfectly possible that both variation and change itself (as a result) are neutral: even selection does not necessarily have to select that which is better ‘adapted’. In any case, there are even in biology modes of (apparent) selection that are not (in the Darwinian sense) genuinely ‘selective’ or ‘adaptive’ [...]. All of these possibilities, given the much better understood nature of variation and change in organisms, need to be considered before any claim for ‘function’ can be made for either variation or change. (Lass, 1997, 354)

ould not be

To finish off

- (Recapitulation:) Different (potential) modes of change:
  - Heraclitean: ill-behaved (random-walky) and non-functional
  - Platonic: well-behaved and non-functional
  - Aristotelian: well-behaved and functional (teleological)
- (Conclusion:) Platonic language change is possible
- (Envoi:) It remains to show that it is probable, i.e. that a level of linguistic representation exists on which change can be largely neutral
  - (Wish:) A new kind of marriage of modelling and empirical data collection?
  - (Hope:) What is a mathematical model of language change? A **laboratory** for diachronic linguistics.

Acknowledgements

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References

Blythe, R. A. & Croft, W. (2012). S-curves and the mechanisms of propagation in language change. *Language*, *88*(2), 269–304.

Chomsky, N. (1995). *The minimalist program.* Cambridge, MA: MIT Press.

Fagyal, Z., Swarup, S., Escobar, A. M., Gasser, L. & Lakkaraju, K. (2010). Centers and peripheries: Network roles in language change. *Lingua*, *120*, 2061–2079.

Ke, J., Gong, T. & Wang, W. S.-Y. (2008). Language change and social networks. *Communications in Computational Physics*, *3*(4), 935–949.

Lass, R. (1997). *Historical linguistics and language change.* Cambridge: Cambridge University Press.